

SWARAJ GUPTA

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TECHNICAL SKILLS

Programming: Java, Python, C++
Database Mgmt: MySQL, MongoDB, MinIO S3, Redis*, HQL*
Big Data: Apache Airflow, NiFi, Hadoop, Hive*, Spark, Kafka, Parquet, Iceberg, Arrow Flight, Dremio, BigQuery, DBT*
ML: TensorFlow, PyTorch, Keras, XGBoost, LightGBM, SVMs, Alibi Detect, Generative AI, Knowledge Graphs*, Exploratory Data Analysis, Classification, Regression, Statistical Models, Predictive Analysis, Tree-Based Models, Reinforcement Learning*, Natural language processing*, A/B Testing
ML Frameworks: MLflow, PyCaret, AutoML, Feast, Milvus, KubeFlow, KServe
Cloud: Kubernetes, Docker, Helm, Istio*, Azure(AZ-900)
DevOps: CKAD, Ansible, Terraform, GitLab CI/CD, Jenkins*
Frameworks: Git, HTTPD, RabbitMQ, Flask, Postman, Prometheus, Grafana, Power BI, SharePoint
General: Data Structure, Algorithm, System Design, Artificial Intelligence*, Agile Development *Elementary Proficiency**

PROFESSIONAL EXPERIENCE

ERICSSON GLOBAL

Software Engineer — Jan'22 - Present

➤ Big Data Solutions:

- Designed, implemented, and optimized scalable distributed ETL/ELT pipelines for 9 telecom customers using PySpark and HDFS. Developed Spark streaming and data migration pipelines between RDBMS, NoSQL, and Cloud (S3).
- Implemented messaging queues and validated data integrity using REST APIs, reducing reprocessing time by 90% and accelerating root cause analysis (RCA). Decreased data load time by 75% through asynchronous multiprocessing.
- Reduced analysis time by 65% by optimizing large datasets with Dremio Reflections. Developed and hosted Flask endpoints to parse and execute queries using a data warehouse engine.

➤ Cloud Native Solutions:

- Orchestrated Directed Acyclic Graphs (DAGs) for Big Data pipelines using KubernetesPodOperator in Airflow, enhancing workflow efficiency by 35%. Ensured seamless operations, comprehensive monitoring, governance, and timely alerts.
- Realized next-generation Cloud-Native Big Data applications, achieving a 100% microservices architecture, higher fault tolerance, scalability and benchmarking 20% better results.
- Implemented secure and efficient cloud-native object storage with MinIO (S3), enhancing data governance, provisioning hybrid storage, and enabling data tiering, which reduced technical debt by 70%.

➤ MLOps Solution:

- Tested and trained Time-Series forecasting models with validation accuracy of up to 75%, and data drift detection improving user experience and RCA for essential business metrics. Supported SMEs with detailed dashboards.
- Engineered machine learning pipelines with Model-as-a-Service, integrating feature selection, transformers, and automated retraining pipelines below 60% accuracy. Ensured continuous monitoring and metrics tracking with MLflow.

ÉTS, Montréal *Evolving Trends in Extended Reality (XR) Softwares*

Research Intern — Summer Jan'21

- Extracted 20,000+ XR software artifacts and technologies via automated API calls from Stack Overflow and GitHub API. Utilized skills in Data Mining, GraphQL, API Rate Limits, and Recommendation Systems for Evolving Software (RSES).

SELECTED PROJECTS

Large Language Model Powered Wikipedia Chat Assistant with Retrieval Augmented Generation

Live App

Key Skills: LLM, RAG, GraphRAG, OpenAI API, Pydantic, Chainlit, Langchain, LlamaIndex, ReAct Prompt Agent

Name Calling Detection using Siamese Convolutional Neural Network

GitHub

Key Skills: Deep Learning, CNN, librosa, Audio Processing (Mel-Spectrograms), Acoustic Word Embedding

Collaborative Filtering Movie Recommendation Engine

Live App

Key Skills: NumPy, pandas, scikit-learn, cosine similarity, Recommendation System, Collaborative Filtering, Streamlit

ACHIEVEMENTS & EDUCATION

- 🌟 **CKAD:** Linux Foundation CNCF Certified Kubernetes Application Developer. 2024
- 🏆 **Samsung Hackathon Winner:** AIR 1st place out of 220+ teams, Samsung R&D Institute, Bengaluru. 2022
- 📄 **MITACS Globalink Research Internship Award:** École de technologie supérieure, Montréal, Canada. 2021
- 📄 **TEDxIIITNayaRaipur Lead Organizer and National Himalayan Trekker YHAI,** Govt. of India. 2021

Certificate	Discipline	Institute	Year	Score
B.Tech	Electronics & Communication	I.I.T. N.R.	2022	82%
Senior Secondary	Science	M.B.V.B.	2018	82%
Secondary	CBSE	M.B.V.B.	2016	10.00